

Getting Started Team Charter

Team name: Kid

Team members: JiaHao Ni, YueShan Li, Yuzaiyang Fan, Ying Xu, Zhenyu Zhang, Zhengxi Chen, Yifan Wu, Xinkai Zhang, Yiming Zhang

1 Project goals and sub-systems

Overall project goal:

Design, construct and test an autonomous vehicle system that can navigate from a specified start position to a designated goal through a road environment containing curved sections and obstacles, while planning a feasible path and avoiding collisions without manual intervention, and drive smoothly while meet the refereed speed.

Sub-systems required to achieve the overall project goal:

1. State estimation and vehicle modelling: Process the provided perception outputs to estimate the vehicle and road state required by downstream planning and control algorithms, and identify/calibrate relevant vehicle parameters.
2. Path planning and obstacle avoidance: Generate a smooth and feasible reference trajectory through the road environment, including appropriate responses to obstacles and road curvature.
3. Baseline trajectory tracking and speed control: Develop and test a reliable baseline controller, such as Pure Pursuit combined with PID/cruise control, for following the generated trajectory.
4. Advanced model-based control: Develop and evaluate an MPC/MPCC-based controller for trajectory tracking, incorporating vehicle dynamics and relevant system constraints.

2 Getting to know each other (i.e., building empathy)

Reminder: these are team building activities that focus on sharing the details that past students have told us are relevant for having a healthy and enjoyable team collaboration.

Background of team members:

- Take turns for each person to describe a bit about themselves, for example: communities that they associate with; extra-curricular projects they are working on; positive and negative experiences of university and of university teamwork.
- **YueShan Li:** I'm currently studying for an MSc in Computer Science, specialising in AI. I'm working on a research project of my own, I'm familiar with python libraries and ai sort of things, able to build fully functional AI systems. Last semester, I completed AI Planning for Autonomy, where I gained experience with planning, Markov Decision Processes (MDPs), and reinforcement learning. I received an H1 in the subject, so I would be comfortable contributing to project work involving AI, planning, or reinforcement learning.

- **Yuzaiyang Fan:** I completed my undergraduate degree in Computing and Software Systems at the University of Melbourne and am currently studying the Master of Information Technology, specialising in Artificial Intelligence. Last semester, I completed AI Planning for Autonomy, where I gained experience with planning, Markov Decision Processes (MDPs), and reinforcement learning. I received an H1 in the subject, so I would be comfortable contributing to project work involving AI, planning, or reinforcement learning.
- **Yifan Wu:** I'm currently studying a Master of Mechatronics Engineering at the University of Melbourne. My background is mainly in robotics, autonomous systems, and control. I have experience working on autonomous driving control algorithms, including vehicle dynamics modelling, trajectory validation, parameter identification, and robustness evaluation. I have also worked on robot learning and Sim2Real deployment projects, where I developed simulation environments, trained policies, and transferred them to physical robots.
- **Xinkai Zhang:** I am a Mechatronics Engineering master's student at the University of Melbourne with a background in artificial intelligence, robotics, and autonomous driving. My experience includes robot control, motion planning, trajectory optimization, sensor fusion, and integrating algorithms with physical systems. I have worked on vehicle-mounted drone tracking, autonomous UAV reconstruction, and mobile robot path planning using C++, Python, and ROS. In this project, I will primarily contribute to robot control, motion planning, simulation, testing, and related system integration.
- **Yiming Zhang:** I'm currently studying a Master of Mechatronics Engineering at the University of Melbourne. My background is mainly in robotics, control, embedded systems, and state estimation. I have experience with robot control, system modelling, and embedded development. I have worked on an ROV project (attitude estimation and sensor fusion). I'm familiar with MATLAB/Simulink, Python, C/C++.
- **Zhenyu Zhang:** I'm currently studying a Master of Mechatronics Engineering at the University of Melbourne, with a background in mechanical engineering, robotics, and control. I have experience in system modelling, motion control, ROS 2, and reinforcement learning. I have worked on robotic arm projects involving kinematics, model-based control, robot learning policies, Gazebo simulation, and physical robot implementation. I'm comfortable with Python, C++, MATLAB/Simulink, ROS 2, and Gazebo.
- **Zhengxi Chen:** I'm currently studying a Master of Mechatronics Engineering at the University of Melbourne, with an undergraduate background in Mechatronics Engineering. I have experience in robot control, embedded systems, mechanical design, and dynamics modelling, including ROS2 and MATLAB simulations. I'm familiar with C/C++, Python, MATLAB/Simulink, and SolidWorks. I'm happy to contribute to modeling, programming, and testing.
- **Jiahao Ni:** I am a Master of IT first year student specializing in AI. My background is mainly software development, artificial intelligence, and machines learning. I'm able to use Python, Java, Net logo, and happy to expand my toolbox by learning new libraries.
- **Ying Xu:** I'm currently studying a Master of Information Technology at the University of Melbourne, with a focus on AI. I'm familiar with Python and have experience with machine learning, computer vision, and simulation-related coursework. I'm happy to contribute to coding, testing, and AI-related tasks in the project.

Roleplay a breach of trust and/or expectations:

- As a team, discuss possible scenarios and pick one, for example:
 - A team member inappropriately uses GenAI resulting in fake references being added to the team report.
 - A team member stops responding to messages for a whole week without apparent reason.
 - A team member takes over and re-does the work of another team member because they consider the work of that other team member to be sub-standard and harmful to their final grade.
- Split your team into two sub-teams, one on each side of the breach.
- Each sub-team discusses for the side they represent:
 - **How** they would behave; and
 - **Why** they would behave that way.
- Share back with each other and discuss.
- Incorrect GenAI result:
Immediately flag any problematic GenAI content and references and suspend their use.

The team member who originally used GenAI is responsible for re-verifying all relevant materials, whilst other members cross-check key references.

All literature ultimately used must be verifiable via Google Scholar, the university's databases, or the publisher's website.

If the original source of GenAI-generated content cannot be found, it must be deleted immediately; do not attempt to 'make up a similar reference'.

The team subsequently agreed that GenAI may be used to assist with searching and organizing information but must not be relied upon directly as a source of facts or references.

- Negative attitude towards work and communication:
First, a team member should contact the missing member privately to check whether there are any special circumstances.

If there is still no response, the team should reassign their key tasks to prevent the project from stalling.

Upon the member's return, they are not required to publicly explain their personal circumstances, but they must reconfirm how much work they are able to take on.

Thereafter, establish minimum communication rules, such as: even if unable to work, a brief message should be sent within 24–48 hours explaining that the work cannot be completed as planned.

If there is another prolonged period of non-communication, formally notify the tutor or supervisor.

- Redo others work:

You should not directly delete or completely overwrite the work of the original contributor.

If you believe the quality of a contributor's work is substandard, first highlight the specific issues via comments or tracked changes.

Give the original author an opportunity to make corrections, and set a clear internal deadline.

If time is too tight, two people can work on the revisions together, rather than one person taking over entirely.

If a substantial rewrite is ultimately necessary, retain the original version and clearly explain why the changes were made.

Going forward, the team agrees that any major rewriting of another person's work must be notified to the original author in advance.

Behavioural style of each individual:

- It can be beneficial to explicitly discuss styles that are likely to be implicitly observed.
- Summary table: <complete for each team member with a "yes" or "no" in each cell, and add any extra styles >

Style	YueShan Li	Yuzaiyang Fan	Zhengxi Chen	YingXu	Zhenyu Zhang	Yiming Zhang	Xinkai Zhang	Yifan Wu	Jia Hao Ni
Often works late at night.	No	No	No	Yes	Yes	no	no	no	yes
Prefers not to work on the weekend.	No	No	Yes	No	NO	no	no	no	no
Is comfortable delaying work on assignments and cramming just before the due date.	No	No	No	No	NO	no	no	no	no
Is stressed-out by leave assignment work until the last days.	No	Yes	Yes	No	Yes	yes	yes	yes	no
Prefers to avoid coming into campus.	Yes	No	Yes	No	NO	no	no	no	yes

Likes to write things down.	No	No	Yes	Yes	NO	no	no	yes	yes
Tend to talk about topics but can't be bothered to write things down.	Yes	Yes	Yes	No	Yes	no	no	no	no
Regularly prioritizes working on extra-curricular projects instead of subjects.	Yes	Yes	Yes	No	Yes	no	no	no	no
Prefers to avoid using GenAI	No	No	No	No	No	no	no	no	no
Uses GenAI for everything	No	No	No	No	No	No	No	No	No
<more styles>									

Familiarity and ability with tools:

Summary table:

- <fill-in the table for each team member and add any extra tools you think are relevant>
- <below the table, add any important notes or skill-development actions that arise from discussing these tools, including whether team members can support each other with skill development>
- Scale: <adapt this scale if you wish>
 - 0 = never used
 - 1 = have used but not confident
 - 2 = confident that can quickly figure out any new aspects about the tool
 - 3 = expert

Tool	Jia Hao Ni	YueShan Li	Yuzaiyang Fan	Zhengxi Chen	Ying Xu	Zhenyu Zhang	Yiming Zhang	Xinkai Zhang	Yifan Wu
MS Teams	1	1	1	1	2	1	1	2	2
git	2	2	2	1	2	1	1	2	1
ssh	1	2	2	0	1	0	1	1	1
Trello (or similar)	0	0	0	0	0	0	0	0	0
Python	3	3	3	2	2	2	1	0	1
ROS	0	0	0	2	0	2	1	2	1
Latex	2	1	1	2	2	2	1	1	1

GenAI for coding	2	3	3	2	3	3	2	1	2
<more tools>									

Skill development and support:

Team members with stronger Python and GenAI coding experience can support others with software development and debugging. Members with more experience in ROS, control, and robotics can assist teammates who are less familiar with these tools. The team will also share knowledge of Git, SSH, and other project tools as needed during development.

3 Individual availability and responsibilities

Availability and other major workloads of each team member:

- Examples for how to describe this: number of other subjects; number of time-intensive subjects; extra-curricular projects that you passionately work on; paid work; travel plans.
- Yuzaiyang Fan: Enrolled in three other subjects this semester — Computer Vision (COMP90086), Machine Learning Applications for Health (COMP90089), and Software Processes and Management (SWEN90016). All three subjects include group projects with written reports, so availability may be more limited around major project and report deadlines. Otherwise, generally available for regular team meetings and project work.
- Ying Xu: Enrolled in two other subjects this semester — Computer Vision (COMP90086), Computational Modeling and Simulation (COMP90083). All three subjects include group projects with written reports, so availability may be more limited around major project and report deadlines. Otherwise, generally available for regular team meetings and project work.
- Jiahao Ni: Enrolled in two other subjects this semester Computer Vision (COMP90086), Computational Modelling and Simulation (COMP90083). Some effort needs to be assigned to prepare for final exam and group project works of these subjects.
- YueShan Li: Enrolled in three other subjects this semester — Computer Vision (COMP90086), Computational Modeling and Simulation (COMP90083), and Statistical Machine Learning (COMP90053). All three subjects include group projects with written reports, so availability may be more limited around major project and report deadlines. Otherwise, generally available for regular team meetings and project work.
- Yifan Wu: Enrolled in Mechatronics Engineering and has previous experience in robotics and autonomous driving, mainly focusing on control, motion planning, and system integration. Will primarily contribute to robot control, motion planning, simulation, and related system integration tasks in the project. Generally available for regular team meetings and project work, with availability potentially reduced around major assignment and assessment deadlines.

- Xinkai Zhang: I am generally available for regular team meetings and project work, although my availability may be reduced around major assignment and assessment deadlines.
- Zhenyu Zhang: Enrolled in two other subjects this semester, including advance control systems and a Capstone project involving robotics, robot control, ROS, and simulation. Both subjects involve time-intensive project work, so availability may be more limited around major projects and report deadlines. Otherwise, generally available for regular team meetings and project work.
- Zhengxi Chen: Enrolled in three other subjects this semester, including Advanced Dynamics, AI for Robotics, and Java programming, with time needed for assignments, coding, and exam preparation. I will schedule regular project work and team meetings around these commitments and communicate any availability changes in advance.
- Yiming Zhang: Enrolled in three other subjects this semester — Embedded System Design (ELEN90066), Advanced Dynamics (MCEN90041), and Interdisciplinary Design for Engineering (ENGR90051). Workload may be heavier around major assignment and project deadlines. I do not have any major travel plans this semester and am generally more available for team meetings and project work on weekdays.

Nominated team service roles:

- <you should add and subtract roles to match what is appropriate for your team as the project progresses>
- Scheduling and coordinator of team meetings: <Ying Xu>
- Submitting Teamwork assignments on Canvas: <Jiahao Ni>
- Ensuring that there is a written record of key decisions: <Yueshan Li>
- Steward of good record keeping: <Ying Xu>
 - Note: This role reviews where action is needed and informs the respective team member of the action they need to take.
- Steward of coding conventions and clean-up: <Yifan Wu>
 - Note: This role reviews where action is needed and informs the respective team member of the action they need to take.
- Progress check-ins between meetings: <Yuzaiyang Fan>
- Personal check-ins between meetings: <Xinkai Zhang>
- **Group presentation (guide) + road and navigation task design or object detection:**
Jiahao Ni